NEXVIA AEGIS V9-M4 Independent Statistical Audit

Phase 7: Full Regression Verification, Diff Integrity, and Finding Severity Register

Autonomous Verification Subsystem (Google DeepMind Antigravity)

Git Branch: v9.0-runtime-assurance · Commit: 083ade9 · Date: October 25, 2026

Abstract

This audit report details the comprehensive regression testing, source code diff audit, and integrated findings severity register for Milestone V9-M4 of the NEXVIA AEGIS runtime assurance architecture. We record empirical results from the focused 14-test synthetic preflight test suite and the complete 125-test repository regression suite (`PYTHONPATH=src uv run pytest -q`), confirming zero test failures across all components. We inspect the repository Git working tree to ensure that no modifications have compromised the frozen selected policy, controller dynamics, sensor models, calibration routines, or historical calibration/validation artifacts. Finally, we consolidate all audit findings into a formal, prioritized Finding Severity Register with concrete remediation specifications.

Keywords: Regression Testing, Git Diff Verification, Software Quality Assurance, Finding Severity Register, Invariant Protection, Holdout Provenance.

1 Executive Summary & Audit Mandate

Phase 7 serves as the synthesizing audit phase:

1. **Empirical Test Verification:** Execute the full automated regression suite to ensure algorithmic stability and prevent regression errors.
2. **Repository Diff & Invariant Audit:** Inspect all tracked files to verify that frozen policies, controller parameters, sensor models, and historical datasets have not been altered.
3. **Finding Severity Categorization:** Formalize and rank all discovered findings according to standardized research quality criteria (Critical, High, Medium, Low).

2 Empirical Regression Test Outcomes

2.1 Focused Synthetic Preflight Suite

We executed the 14 targeted preflight test cases in `tests/test_v9_holdout_preflight.py`:

- **Execution Command:** `PYTHONPATH=src uv run pytest -v tests/test_v9_holdout_preflight.py`
- **Recorded Result:** 14 passed in 0.99 seconds (Exit Code: 0).
- **Coverage:** Tested outcome algebra invariants, manifest and policy corruption detection, configuration tampering rejection, undefined selective risk handling, isolated mixture and scenario evaluation, single-shot overwrite guards, fail-closed execution barriers, seed boundary checks, and read-only CLI status.

2.2 Full Repository Regression Suite

We executed the full automated regression suite across the entire codebase:

- **Execution Command:** `PYTHONPATH=src uv run pytest -q`
- **Recorded Result:** 125 passed in 30.72 seconds (Exit Code: 0).
- **Failure Count:** 0 failures, 0 errors, 0 warnings.
- **Component Breakdown:**
  - Core kinematic controllers and MPC baselines: Verified.
  - Multi-hypothesis Bayesian diagnosis and Ville martingales: Verified.
  - LTT calibration and fixed-sequence testing algorithms: Verified.
  - V9-M3 operational validation regression baseline: Verified.
  - V9-M4 confirmatory evaluation harness preflight: Verified.

3 Repository Working-Tree & Invariant Audit

We audited the repository working tree against starting commit `083ade9`:

- **Policy Invariance:** The calibrated policy `results/v9/cal_r2/selected_policy.json` remains identical to commit `150629a` (raw SHA-256: `5a89d6061b379f1b34ab47be0deef8d63180d3e4c21d18c478bc7e451f5c84e2`).
- **Controller Dynamics & Sensor Models:** No modifications in `src/aegis/v9/` or `src/aegis/next/`.
- **Calibration Engine:** `src/aegis/v9/risk_statistics.py` and `experiments/v9_calibrate_ltt.py` unmodified.
- **Historical Validation Artifacts:** `results/v9/val_r1/` artifacts remain bit-for-bit identical.
- **Holdout Data Integrity:** Confirmed that no holdout manifests or output folders have been created.

4 Consolidated Finding Severity Register

Table 1 details all verified findings identified across Phases 1 through 7 of the independent statistical audit.

ID	Severity	Title / Area	Finding Description & Practical Impact	Remediation Requirement
M4-F01	Low	Protocol Evolution in Hypothesis Indexing (H_2 vs H_4)	Early draft narrative labeled H_2 as core-10 mixture risk and H_4 as compound McNemar. Finalized M4 review formalized H_2 as unconditional FS reduction and H_4 as coverage floor. Harness correctly follows finalized M4 review.	Fully reconciled in audit documentation.
M4-F02	**High**	Estimand Inversion in Berger Intersection-Union Test (H_5)	Protocol specifies IUT over unconditional incidence $R_{\text{inc},s}$ on $N_s = 120$. Harness evaluates selective risk on M_s, setting $p = 1.0$ when $M_s = 0$. This penalizes safe abstention in catastrophic fault scenarios and guarantees IUT failure.	Update IUT p-value to $p_{\text{inc},s} = \text{binom.cdf}(fs, n_s, 0.05)$ prior to execution.
M4-F03	Medium	Implicit Key Ordering in Paired Comparisons (H_2, H_3)	Paired loops in lines 555–559 and 755–760 use `zip()` without asserting matching seeds and scenarios. If record arrays are permuted, mismatched episodes are paired silently.	Add explicit assertions: `assert r["seed"] == v["seed"] and r["scenario"] == v["scenario"]`.

Table 1: Consolidated finding severity register for Milestone V9-M4 preflight audit.

5 Detailed Analysis of Finding Severity

1. **Critical Findings (Count: 0):** No critical defects were identified. The execution guards strictly fail closed, holdout partitions remain completely untouched, and no unauthorized live execution has occurred.
2. **High Severity (Count: 1 — Finding M4-F02):** Evaluating the Berger IUT on selective decisions with a fallback of $p = 1.0$ when $M_s = 0$ is a material statistical discrepancy. Because runtime assurance is designed to abstain in catastrophic fault scenarios, this implementation defect would cause the confirmatory evaluation to falsely reject safety (H_5) even if the policy operates flawlessly with zero false successes. Remediation before production execution is mandatory.
3. **Medium Severity (Count: 1 — Finding M4-F03):** Relying on `zip()` for matched-pair comparisons is an unnecessary operational risk that can be eliminated by adding explicit key assertions.
4. **Low Severity (Count: 1 — Finding M4-F01):** Historical documentation evolution, fully resolved by audit Phase 1.

6 Conclusion

Phase 7 confirms 100% regression pass rates across all 125 test suites. The finding severity register provides an actionable, transparent roadmap for pre-execution readiness.
Phase 7 Recommendation: `AUDIT_PHASE_7_PASS_WITH_QUALIFICATIONS`

References

[1] IEEE Standard for Software Quality Assurance Processes, *IEEE Std 730-2014*, IEEE, 2014.

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